

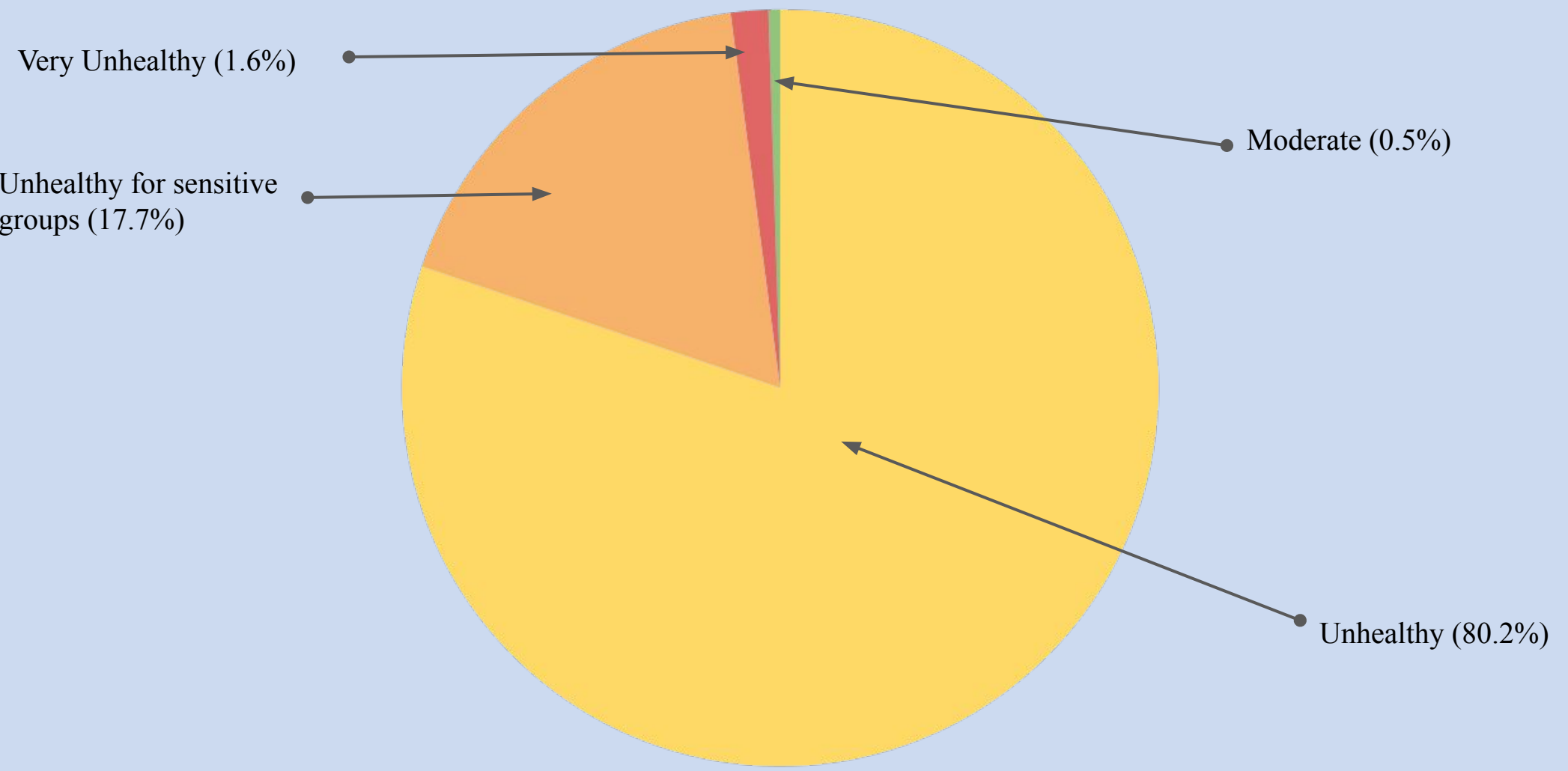
Acknowledgments

A special thanks to Mr. Craig Queenan and Dr. Dina Ellsworth for their invaluable guidance and support. I am also grateful for NJAS and AJAS for the opportunity to share my findings.

Rationale

Over **91% of the global population** resides in regions where air quality surpasses the limits set by the World Health Organization, highlighting a pressing concern for public health. The most prevalent pollutants contributing to this issue include particulate matter (PM), nitrogen dioxide (NO2), sulfur dioxide (SO2), ozone (O3), carbon monoxide (CO), and lead, with fine particulate matter (PM2.5) being the most critical due to its significant health implications. PM2.5 is particularly notorious for its link to a host of health problems, such as respiratory diseases, cardiovascular issues, and premature mortality, posing a major challenge in urban environments.

Fig 1. - Mumbai NowCast AQI Index (2023)



Currently, the Air Quality Index (AQI) provides a standardized measure of air pollution levels by quantifying concentrations of these pollutants. However, this **index falls short by not considering other influential factors**, including meteorological data like temperature, humidity, and wind, as well as traffic and urban activities that significantly contribute to air pollution levels. Moreover, there is a **notable absence** of comprehensive methods to **assess the effectiveness of air pollution countermeasures** through adaptable and feature-extractive neural networks.

Solution

The neural network will **address current methods failing to incorporate environmental factors** and will be able to use the data to identify which countermeasures will be most effective.

This innovative approach aims to transcend the limitations of the conventional Air Quality Index (AQI) by integrating a broader spectrum of influential factors beyond the standard pollutant concentrations. By incorporating meteorological data, such as temperature, humidity, and wind speed, alongside variables representing traffic flow and urban activities, the proposed neural network **endeavors to provide a more accurate and holistic assessment of air quality**.

The significance of this solution lies not only in its capacity to offer real-time evaluations of air quality but also in its potential to **predict future conditions**, thereby facilitating preemptive measures against deterioration. This model sets the stage for a comparative analysis of air quality before and after the implementation of various pollution countermeasures. It will do so by identifying any parameters a countermeasure will affect and then predict AQI changes based on future forecasted values. Such a solution is instrumental in **identifying the most effective strategies for mitigating air pollution** in specific locales, enabling policymakers and air pollution analysts to allocate resources and efforts more efficiently.

Optimizing Air Quality: A Deep Learning Approach Integrating Various Data for Effective Countermeasure Evaluation

Rohit Kottomtharayil - High Technology High School, Lincroft, NJ

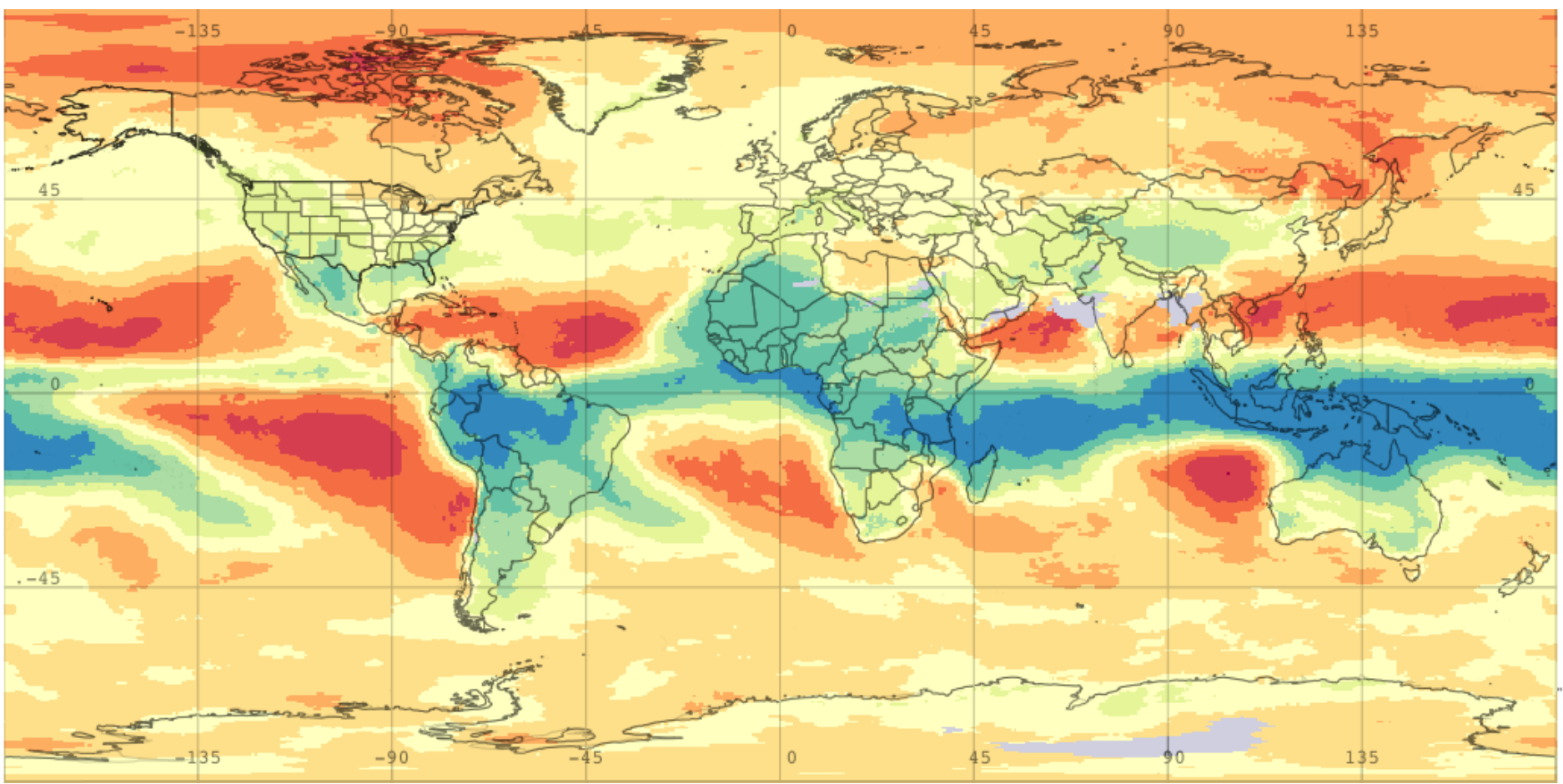


Fig 2. - Average Cloud Top Pressure (Jan. 2024 - Feb. 2024)

Feature Selection

In order to create a model with the highest accuracy, features were plotted against each other to gain an understanding of the correlation. An algorithm known as Extra Tree Regressor was used to extract the top ten influential features for the dataset; this is achieved by using the feature importance property and assigning a score to each feature. Some of the most important meteorological features were average wind speed, average relative humidity, maximum sustained wind speed and average visibility.

Compiling Data

Raw Datasets

- United States Environmental Protection Agency (EPA)
 - Pollutant Data: criteria gases, particulates, toxics, and air quality index
- U.S. Department of Transportation Open Data Catalog
 - Traffic Data: active work zones, site analytics, peak traffic hours, highly populated transportation areas
 - For cities outside of the U.S., traffic data was collected from ministries of transport or private companies.

Meteorological Data

- Sourced from NASA Giovanni's datasets, specifically **MERRA-2**
- Developed a program to convert data from heatmaps to dataframe
 1. Define time frame and geographic coordinates
 2. Select meteorological parameters
 3. Generates URLs to extract data
 4. Interprets and converts data
 5. Organizes processed data frame in CSV file

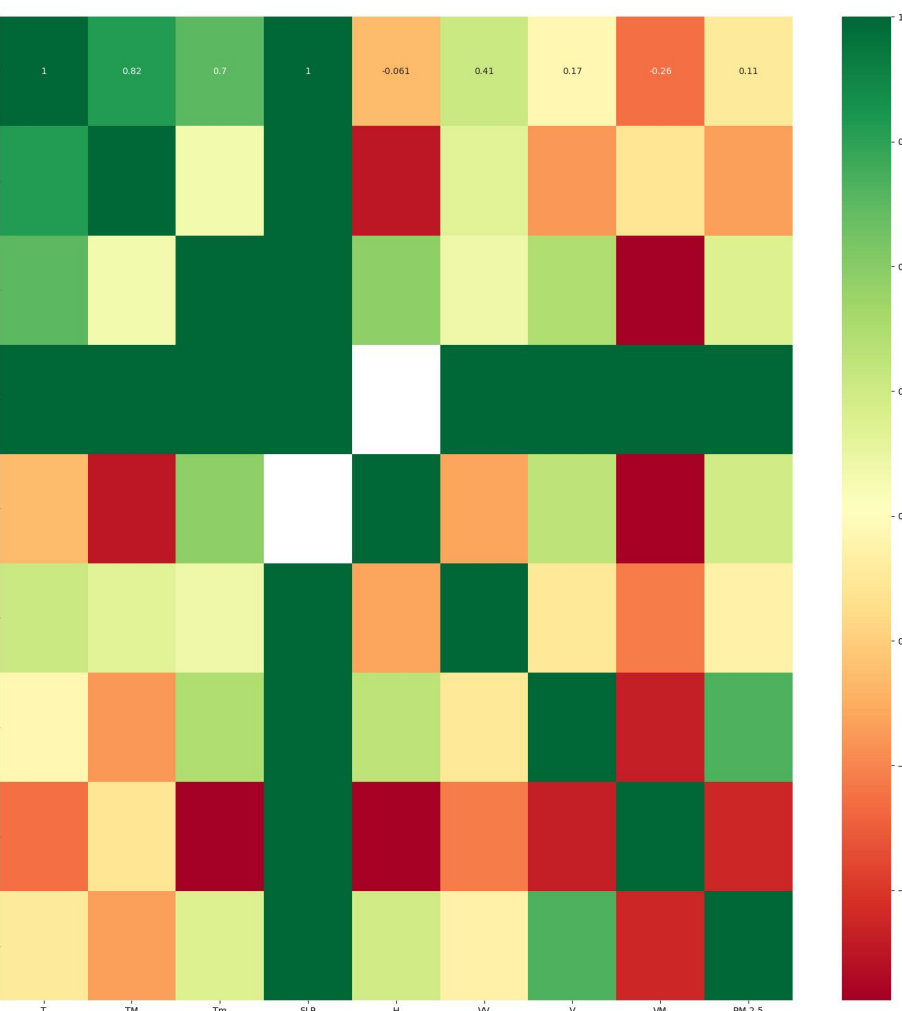


Fig 3. - Feature Selection Heatmap

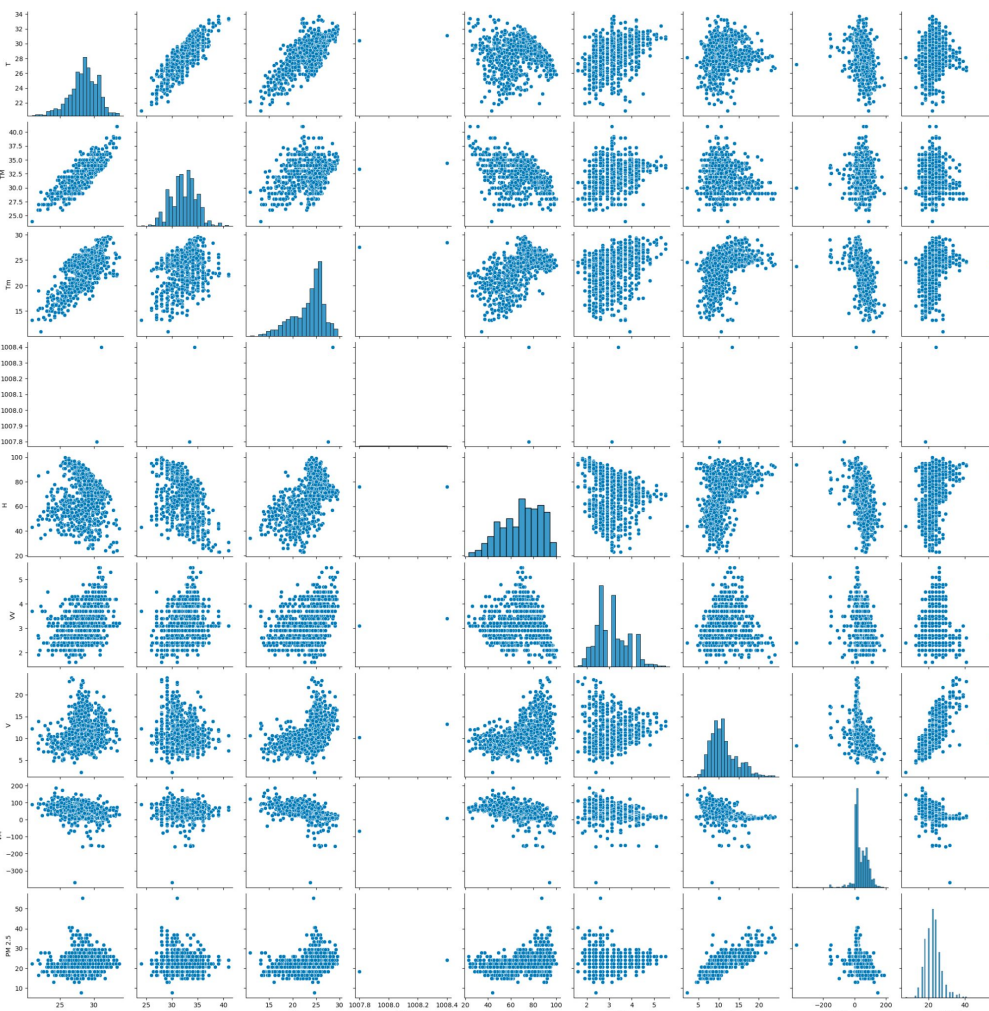


Fig 4. - Feature Selection Pairplot

LSTM Model

In this specific implementation, the LSTM model takes a sequence of observations as input and predicts the AQI value for the next time step. The model is defined with an LSTM layer consisting of **100** units. The layer at the end outputs the predicted AQI value. The model is trained with a significant number of epochs (**500**), indicating an extensive learning process, and utilizes a batch size of **32**, which determines how many samples are worked through before the model's internal parameters are updated. Each epoch contained eight steps, each of which took approximately 4ms. In total, the time taken to run this model was 32 seconds, which is significantly faster than the other models. These given specifications achieved a testing RMSE which is significantly lower compared to the conventional models with a score of **0.26610321114602065**, showing how highly accurate this model is compared to the others.

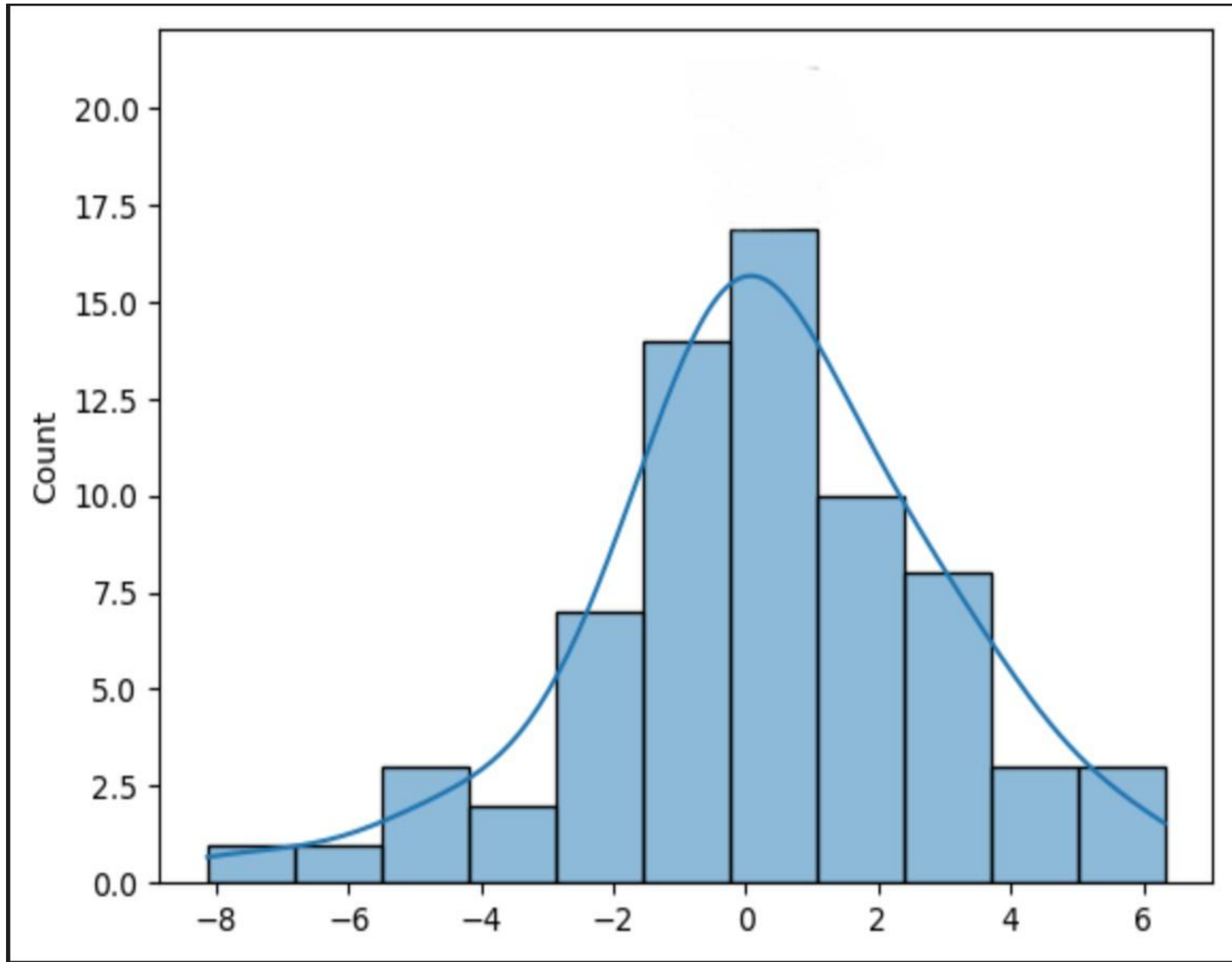


Fig 5. - LSTM Sample Regression (Mumbai 2023)

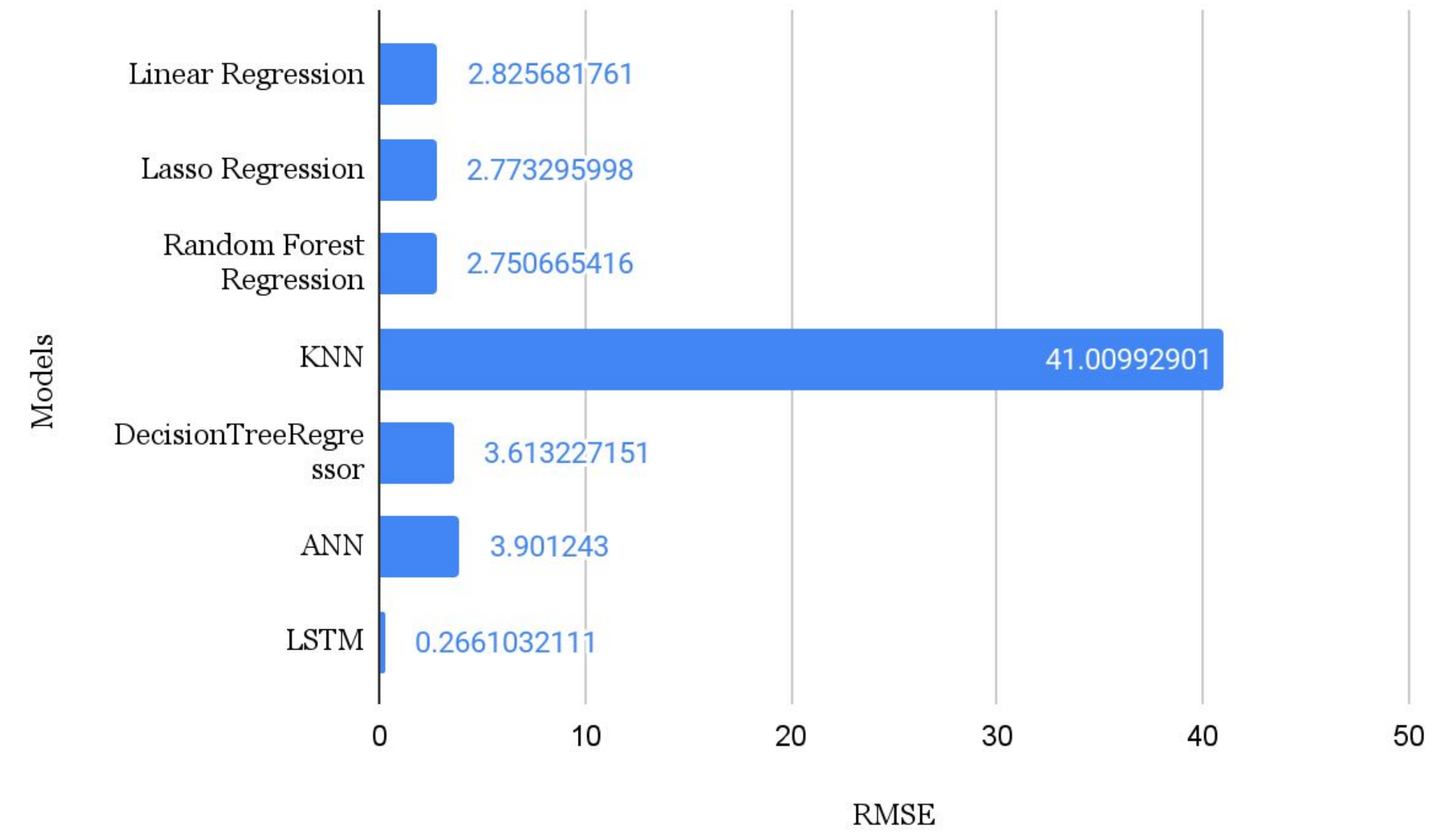


Fig 6. - Regression Model RMSE (Mumbai 2019-2023)

Countermeasure Evaluation

After the point of creating a highly accurate model to predict AQI levels into the future, the model can then be used to gain an understanding of which countermeasures are most effective. This can be done by analyzing the percent decrease resulted by each historical countermeasure and what parameters were involved the most with each countermeasure.

Table 1 - Mumbai AQI Countermeasure Impact

Construction Dust Control (2016)	-25.939%
Promoting Public Transport (2019)	-9.248%
Industrial Emission Control (2019)	-11.432%
Green Initiatives (2022)	-2.142%
Electric Vehicles (2023)	8.3335%

The percent decrease in AQI based on these different countermeasures was calculated by interpreting what parameters the countermeasures would impact and then apply the LSTM regression to calculate the percent decrease. Mumbai was chosen as the city to demonstrate the countermeasure evaluation due to several meteorological factors given the coastal nature of the city, the poor air quality levels, and the various relief actions deployed in the past five years.

Table 2 - Mumbai 2025 AQI Prediction

Promoting Public Transport - Ex. Affected Parameters: Traffic peak hours, traffic concentration, criteria gas levels	-11.540%
Traffic Management - Ex. Affected Parameters: Traffic peak hours, traffic concentration, criteria gas levels	0.022%
Industrial Emission Control - Ex. Affected parameters: Visibility, criteria gases, toxics,	-11.502%

Discussion

In the example countermeasure evaluation above, it is evident the power that this LSTM model holds in terms of interpreting how effective countermeasures can be for scientists.

If utilized properly, the model can give insight into what time period would be the most effective for a certain countermeasures given historical and upcoming meteorological, traffic and pollutant data.

Furthermore, this LSTM model optimizes the time complexity required to keep the runtime under one minute. This allows the user to forecast given parameter values in times much faster than given models that rely on basic forms of regression.

This tool can be incredibly useful in cities where air quality is incredibly poor, such as Mumbai, to help assist air pollution analysts decide what countermeasure to implement based on the given meteorological, traffic, and pollutant data.

Future Studies

The future goal of this project is to be able to recommend a countermeasure for any given area based on historical data of countermeasures and the current environmental state of the area. For example, for high-traffic zones such as Chicago, the model would be able to recommend countermeasures of traffic management and suggest the timeframe that it should be conducted. This would automate the process for users and provide even more efficient insight for air pollution analysts.